

MANU PRASAD

Senior Mechanical Design Engineer

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PROFESSIONAL PROFILE

Senior Mechanical Design Engineer with 6 years of experience taking electromechanical and robotic systems from concept through production including autonomous ground vehicles (AGV), UAVs, underwater robotics, and NABL-certified medical devices. Hands-on strength in first-principles design calculations (drivetrain and motor sizing, traction and load analysis, tolerance stack-up), Innovative and detailed design, and DFMEA/DFM/DFA-driven development. Co-inventor of a patented handheld dynamometer (Patent No. 356182-001); has led full-lifecycle development of 17 products, 7 currently in market

SELECTED WORKS

- **AGV Mechanical Design (Neuralzome Cybernetics):** Drivetrain sizing, bin level-change and pick-and-place mechanisms, and component selection for a 150 kg-class indoor/outdoor autonomous ground vehicle. Platform advanced to field trials
- **High-Volume DFMA:** 32.7% parts-count reduction on a production machine while holding structural integrity, enabling a 1,000-unit run with CNC, laser-cut and blow molded parts with precision tolerances
- **FITMUST Dynamometer (Patented):** India's first NABL-certified handheld dynamometer — end-to-end mechanical design, sensor calibration fixtures, ISO 13485/14971 compliance, and 500-unit mass production with 36% BOM cost reduction

CORE COMPETENCIES

- **Design Calculations & Analysis:** First-principles component sizing | Drivetrain & motor sizing | Load, traction & torque analysis | Tolerance stack-up | FEA (ANSYS) | GD&T | Vibration mitigation
- **CAD & Documentation:** SolidWorks (expert) | Fusion360 | Parametric modeling | Manufacturing drawings (70+ released) | BOMs | Sub-assembly procedures & technical documentation
- **Manufacturing & Prototyping:** CNC machining | Sheet metal | Welded & fabricated structures | Injection molding & soft molding | Advanced 3D printing (PA6-CF, PETG) | Precision tolerances ($<20\text{ }\mu\text{m}$) | Production scaling (1,000+ units)
- **Quality, Validation & Compliance:** DFMEA/PFMEA | Prototyping, testing & validation | Custom test fixtures (thrust stand, 10-bar pressure chamber) | IP68 sealing | ISO 13485 / ISO 14971 | Six Sigma & root-cause analysis
- **Leadership & Collaboration:** Cross-functional teams (mechanical, electrical, firmware, quality) | Vendor & supply-chain management | Agile development | Value Engineering (VE/VA)

PROFESSIONAL EXPERIENCE

Head of Mechanical Systems — Stratofirma Autonomy Labs

Oct 2025 – Present | Bangalore

Overwatch (Hexacopter)

- Spearheaded the mechanical architecture of a high-performance hexacopter, achieving a 62% increase in torsional rigidity and 27% weight reduction through a square-frame design. Defined a DFM-focused, field-repairable architecture with a faceted avionics spine, square carbon-fiber arms with machined clamp hubs, and a rectangular load-frame landing gear
- Mitigated dynamic flight vibrations by selecting and engineering specialized elastomeric mounts at critical arm-to-body and landing-gear interfaces

Outrider (Mini Drone)

- Led electromechanical packaging of a dense avionics stack (Nvidia compute board, H7 flight controller, GPS, Wi-Fi module, spatial camera) and resolved severe EMI/RF interference through subsystem compartmentalization and optimized spatial layout with zero signal degradation

Prahar (Micro Drone)

- Engineered a fully IP-rated micro drone for rugged deployments innovated a conical airframe (patent pending) that autonomously mates with a custom ceiling-mounted docking station

Consultant — Mechanical Design & Manufacturing

June 2023 – Oct 2025 | Bangalore

Neuralzome Cybernetics | Andromeda Maritime Solutions | Yangtso Four Labs | Velectron Labs | Techswarm

Mobile Robotics — AGV Platform (Neuralzome Cybernetics)

- Performed first-principles drivetrain sizing for a 150 kg-class indoor/outdoor AGV carrying 40 kg bins: traction analysis for 5° grade traversal on loose soil ($\mu = 0.6$, 1.5 FoS) yielding a 1,568 N tractive requirement; specified 24 V, 400 W BLDC drive motors with 30:1 reduction delivering 30 N·m per wheel at ~100 RPM (0.8 m/s design speed)
- Engineered the bin level-change mechanism with 8 m vertical travel handling 40 kg crates: sized brake motor torque (39.4 N·m) from combined dynamic, gravitational, and rail-friction loads; specified encoder feedback, dual limit switches, and cable drag chain routing

- Sized the telescopic pick-and-place arm drive (60 kg horizontal payload, 0.7 m stroke in 1.4 s) and selected a NEMA23 planetary-gear BLDC (100 W, 3.5 N·m) with servo-actuated retention flaps
- Owned component selection and sourcing across drive, actuation, and sensing subsystems (3D LiDAR, stereo depth cameras, hub drives, 30 kg-rated casters); platform validated bin handling in confined indoor aisles and progressed to field trials

Underwater & UAV Systems

- Designed and developed an underwater rover capable of 4 knots with 10 kg payload; engineered IP68 sealing systems for a UAV rated for 100 m underwater operation
- Engineered a thrust stand (up to 50 kg force) and a 10-bar pressure chamber for component qualification and sealing validation. Design and tested penetrators and pressure valves which are integral part of the ROV

Production-Ready Mechanical Systems

- Implemented DFMA methodology achieving 32.7% parts reduction while maintaining structural integrity, accelerating assembly for a 1,000-unit production run
- Designed large-scale machine structures (8 ft and 26 ft) engineered for 80 kg and 200 kg loads with validated structural performance
- Designed, manufactured, and assembled 35+ sheet-metal components with complete documentation, sub-assembly procedures, and tolerance stack-up analysis; released 70+ GD&T-compliant drawings for sheet-metal, CNC, welded, and molded parts
- Optimized components for CNC and laser cutting to tolerances under 20 microns; engineered a sensor-free spring-based ball-dispensing mechanism ensuring reliable single-ball operation

Electromechanical Product Development

- Designed an ergonomic medical wearable (50 mm × 30 mm) through rapid prototyping with strategically positioned sensor integration; engineered a test apparatus simulating human lung conditions with controlled pressure drop and 1,000 cm³/s smoke extraction

Chief of Product — Ashva Wearable Technologies

March 2020 – April 2023 | Bangalore

New Product Development

- Led end-to-end design and market launch of 3 medical devices (2 B2B, 1 B2C) within 2 years, deployed across 30+ hospitals
- Co-invented FITMUST, India's first NABL-certified handheld dynamometer (Patent No. 356182-001); designed the mechanical system around the load-sensing chain and engineered custom fixtures for system/subsystem testing, sensor calibration, and load validation
- Designed PCB layout for seamless module assembly with dedicated calibration and test boards; facilitated firmware development for testing, calibration, and core functionality

Manufacturing & Quality

- Spearheaded mass production of 500 units, cutting the production timeline by 4 weeks; applied VE/VA to reduce BOM cost by 36% through design optimization
- Ensured ISO 13485 / ISO 14971 compliance with active contribution to DFMEA, PFMEA, and risk mitigation; applied Six Sigma tools (Ishikawa) for root-cause analysis

Leadership

- Led a cross-functional team of 7 engineers through iterative launches; managed 5+ vendor partnerships spanning hardware, firmware, and application development

Freelance Mechanical Engineer — DrinkBarty | Techlytx

Aug 2019 – Feb 2020 | Kollam

- Designed a cocktail-making machine with a Keurig-style custom mixing-cup delivery system; engineered an omnidirectional robot with OpenCV pathfinding on Raspberry Pi + Arduino, incorporating slider-crank mechanisms and pneumatic linkages

EDUCATION

B.Tech, Mechanical Engineering — TKM College of Engineering, Kollam

2015 – 2019